Weighted one-dimensional power substitution

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Abstract

Step 1 of the general-d equal-ratio monomial milestone (Astra #14). For an arbitrary measurable $g\colon \mathbb{R} \to [0, \infty]$,

$$\int_{(0,1]} u^h \, g(u^{2k}) \, du = \frac{1}{2k} \int_{(0,1]} t^{\lambda - 1} g(t) \, dt, \qquad \lambda = \frac{h+1}{2k},$$

in Lebesgue-integral form (`lintegral_weighted_power_subst`), so no integrability is needed and the identity can be iterated inside Tonelli arguments. It rests on the change of variables $t = u^p$ on $(0, 1]$ (`lintegral_Ioc_rpow_subst`), which maps the interval onto itself (`image_rpow_Ioc`), via Mathlib's one-dimensional Jacobian formula. Zero `sorry/axiom`.

1 The things formalised

```
theorem image_rpow_Ioc (p) (hp : 0 < p) : (fun x :   => x ^ p) '' Ioc 0 1 = Ioc 0 1
theorem lintegral_Ioc_rpow_subst (p) (hp : 0 < p) (g :   → ENNReal) :
      t in Ioc 0 1, g t =    u in Ioc 0 1, ENNReal.ofReal (p * u ^ (p - 1)) * g (u ^ p)
theorem lintegral_weighted_power_subst (h k :  ) (hk : 0 < k) (G :   → ENNReal) :
      u in Ioc 0 1, ENNReal.ofReal (u ^ h) * G (u ^ (2 * k))
      = ENNReal.ofReal (1 / (2 * k))
        *   t in Ioc 0 1, ENNReal.ofReal (t ^ ((h + 1) / (2 * k) - 1)) * G t
```

2 Lean notes

- `MeasureTheory.lintegral_image_eq_lintegral_abs_deriv_mul` (file `JacobianOneDim`) with $s = (0, 1]$, $f = (\cdot)^p$, derivative from `Real.hasDerivAt_rpow_const` and injectivity from `Real.rpow_left_injOn` restricted to nonnegative reals.
- Rewriting the substitution into the right-hand side only needs `conv_rhs`; a bare `rw` hits the left-hand integral first.
- `ENNReal.ofReal_mul` with metavariables rewrites the first product it finds; pass the factors explicitly (`@ENNReal.ofReal_mul (1/p) (p * u^(p-1)) …`) before `ring`, which treats the `ofReal` atoms consistently only if both sides are split identically.